URL vs. URI vs. URN

Understanding the differences between URL, URI, and URN can be crucial in web development and networking. Let's break down these concepts:

URL (Uniform Resource Locator):

- **Definition:** A URL is a specific type of URI that not only identifies a resource on the internet but also provides a method to locate it by describing its primary access mechanism, usually its network location.
- **Components:** It typically includes a protocol (such as HTTP, HTTPS, FTP), domain name, and path, optionally followed by query parameters or a fragment identifier.
- **Example:** `https://www.example.com/path?query=term#section`
- **Key Characteristics:**
  - Specifies how the resource can be accessed (protocol).
  - Includes the location of the resource (like a web address).

URI (Uniform Resource Identifier):

- **Definition:** A URI is a generic term used to identify a resource either by location, name, or both. It serves as a universal identifier for resources on the internet.
- **Scope:** All URLs and URNs are URIs, but not all URIs are URLs or URNs.
- **Example:** A URL `https://www.example.com` is also a URI, and a URN like `urn:isbn:0451450523` (identifying a book by its ISBN) is also a URI.
- **Key Characteristics:**
  - A more general concept than both URL and URN.
  - It can be either a locator (URL), a name (URN), or both.

URN (Uniform Resource Name):

- **Definition:** A URN is a type of URI that names a resource without describing how to locate it. It's used to assign a unique and persistent identifier to a resource.
- **Example:** `urn:isbn:0451450523` uniquely identifies a book using its ISBN, irrespective of where it exists.
- **Key Characteristics:**
  - Provides a unique and persistent identifier.
  - Does not specify a location or method to access the resource.

Summary of Differences:

- **URL:** Specifies both the identity and the location of a resource (How and Where).
- **URI:** A more comprehensive term covering both URLs (identifying and locating) and URNs (just identifying).
- **URN:** Focuses only on uniquely identifying a resource, not on where it is located or how to access it.

In practical terms, when you're browsing the internet, you're mostly dealing with URLs. URIs and URNs come more into play in specific contexts like software development, digital libraries, and systems where unique and persistent identification of a resource is crucial.